

PeerForge — Features and Functionality

Verified against a running instance, 2026-08-05. Anything marked **Partial** or **Known limit** is stated as such rather than rounded up.

At a glance

API surface	107 paths, 121 operations
Data model	39 tables, 11 migrations
Backend tests	440 passing, 35 files
Reviewer lanes	6 distinct roles
Integrity rules	11 (10 enforcing, 1 advisory)
Frontend routes	12

1. Institution, courses and people

A three-level hierarchy: **institution** → **course** → **session**.

Feature	What it does	State
Organisations	An institution with a name, slug and seat plan	□
Courses (workspaces)	Each course is its own workspace with its own sessions	□
Roles	<code>org_admin</code> , <code>professor</code> , <code>ta</code> , <code>student</code> at institution level; the same vocabulary per course	□
Seats	Purchased vs used, with only <code>student</code> billable; pending invitations reserve a seat at invitation time, not on acceptance	□
Invitations	Single and bulk, email-delivered, token-accepted, revocable	□
Cohort view	Per-course roster with progress, scoped to that course	□
Last-admin guard	An organisation cannot be left without an admin, including via invitation acceptance	□

Authorization. Every debate-scoped request resolves the debate's workspace and checks membership across **all** of the caller's workspaces. A professor teaching four courses reaches sessions in all four, not just the one currently selected.

2. Getting a session ready

Feature	What it does	State
Document upload	PDF, DOCX, TXT and more; MIME validated before anything is written	☐
All-or-nothing uploads	A batch is validated in full before existing files are touched — an invalid file changes nothing	☐
Text extraction + OCR	Scanned documents are OCR'd	☐
Chunking	Paragraph-aware, 1000 chars with 200 overlap, sha256 per chunk	☐
Embedding	<code>text-embedding-3-small</code> , 1536 dimensions, into pgvector	☐
Conversational setup	Describe the panel you want in prose; the assistant proposes reviewers, roles and rounds, grounded in your uploaded document	☐
Manual panel building	Pick from 22 curated reviewer templates or define your own	☐
Action item extraction	Pulls actionable items from a transcript with priority and owner	☐

Conversational setup, concretely. You type "**a statistician who won't let a weak design pass, a clinical psychology expert, and an examiner who decides publishability — two rounds**" and get a named panel with distinct role lanes, retrieved passages from your own document informing the proposal. Editable before you apply it.

3. Preflight — reviewers prepare before they speak

Each reviewer produces a **private preparation memo** before the session opens. This is the feature that most distinguishes PeerForge from a group chat with personas.

Step	What happens
Retrieve	Semantic search over the session's chunks, plus any context granted from prior sessions
Research	Live web search on the problem statement (Tavily, 5 sources)
Compose	A role-specific memo: what the work claims, the biggest gap, the hardest questions this reviewer intends to ask
Cite	A required External literature consulted section, each bullet tying a source to the critique with its URL

Measured: 3 of 5 retrieved sources cited, in 9 of 9 memos. Before the citation requirement was made a required output section rather than a trailing rule, it was 0 of 5.

When no web-search key is configured, the panel says so plainly and the prompt explicitly forbids inventing URLs to fill the gap.

4. The review session

Feature	What it does	State
Three-stage turns	Private reasoning → visible response → constitutional validation	☐
Six reviewer lanes	advisor · methodology professor · domain expert · skeptical reviewer · friendly professor · external examiner	☐

Feature	What it does	State
Live transcript	WebSocket-driven, with agent thinking steps visible	☐
Manual advance	Next Turn, or Ctrl/Cmd+Enter	☐
Autonomous mode	The panel runs itself with a configurable delay; pause and resume	☐
Human intervention	Interject mid-session as the researcher or moderator	☐
Session controls	Start · Pause · Resume · +2 Rounds · End	☐
Cross-instance broadcast	Redis pub/sub, so more than one API instance stays consistent	☐

5. Integrity — the part that is actually hard

Language models playing reviewers converge. They restate each other, defer to each other, and cite documents that do not exist. PeerForge measures and resists all three.

Eleven rules, checked every turn

Ten force a constrained regeneration; one is advisory. The full table is in [ARCHITECTURE.md](#).

Citation integrity

Situation	Behaviour
No document submitted	Any page or section reference is fabrication by definition — regenerate, and replace any survivor with [source not provided]
Document submitted	Cited sections and tables are checked against what the document actually contains
Section exists	Passes silently
Section absent, family present	Flagged — the document has sections 1–4, the reviewer cited 9
No locators of that family at all	Unverifiable, not fabricated — extraction may have dropped the structure
External literature	(McKay et al., 2018) is never flagged; a reviewer may legitimately know it

Measured: with three reviewers instructed to cite pages, sections, tables and figures in every paragraph, **0 fabricated locators reached the transcript.**

Retry re-validation

A regenerated message is re-checked and gets at most one further attempt, kept only if it clears more violations than it introduces. This exists because the same class of bug appeared three times: a retry introducing "Figure 3", a retry re-citing a missing section, and a retry opening with deference — each time the rule worked and the output was wrong anyway.

6. Assurance and output

Feature	What it does	State
Transcript	Complete, ordered, append-only	☐
Summary	Grounded in the real transcript; refuses to run on an empty one	☐
Research profile	Problem, claim, methodology, dataset, contribution, limitations, extracted from the materials	☐
Defence questions	15 viva-style questions, each with a source excerpt from the document	☐
Answer evaluation	Submit answers; the assessment updates and records its basis	☐
Readiness assessment	10 scored dimensions with a stated basis (has_profile, has_summary, answer_count, message_count)	☐
Signed certificate	sha256 over scores plus ordered evidence, signed; publicly verifiable	☐
Three-valued verification	VALID · SUPERSEDED · INVALID	☐
Glass-Box provenance	Which chunk a claim rests on, with sha256 re-verified	☐
Quality report	Restatement and opener-template rates, gated	☐

Why SUPERSEDED matters. A certificate whose session has moved on is **authentic but stale** — not a forgery. A two-valued verdict called it `INVALID`, which is a different and much more serious accusation.

7. Interface

Twelve routes: `/` · `/setup` · `/setup/chat` · `/room` · `/history` · `/organization` · `/cohort` · `/progress` · `/settings` · `/operator` · `/onboarding` · `/invite` · `/verify/[id]` · `/architecture`.

Aspect	State
Accessibility linting in CI	☐
Keyboard navigation and screen-reader labels	☐
Dark mode	☐
In-app messaging (no <code>alert()</code>)	☐
Never asks for an API key when the server holds one	☐ enforced by a build-time check

That last one is enforced, not merely intended: `scripts/check-key-gates.js` fails the build if any generation gate depends on a browser-held key. It binds the key to whatever name it is given and flags boolean use on either side of `&&` and `||` — it was rewritten after an earlier version passed while nine real gates existed.

8. Operations

Feature	State
Structured logging with request correlation ids	☐
Database connection pooling	☐

Feature	State
Background workers on three queues	☐
Deployment readiness check	☐
Migration runner covering both migration directories	☐
CORS allowlist	☐
Fail-fast on a placeholder JWT secret when auth is on	☐

9. Known limits

Limit	Detail
Page citations are not verified	391 of 398 live chunks carry no <code>page_num</code> ; checking pages would reject legitimate citations
Differentiation degrades by round two	On short papers with few distinct flaws. The gates catch it; the fix is partial
Prep packs cite 3 of 5 sources	Capped deliberately at two or three bullets — forcing all five produces padding
Tests share the demo database	The suite leaves fixtures behind; seed after testing
Tunnel URLs are ephemeral	The API URL is inlined at build time, so a new one needs a rebuild

10. What is deliberately not built

- **No plagiarism detection.** Different problem, different tooling.
- **No reference-manager integration.** No Zotero or Mendeley sync.
- **No LMS integration.** No Canvas, Moodle or Blackboard connector.
- **No mobile app.** The web app is responsive; there is no native client.
- **No multi-language review.** Prompts and rules are English-only.